

Analytical Questions

① check distance between repetitions

If distance there a common factor

$$\text{Distance} = k \times \text{column_size}$$

* If random $\rightarrow$ Repetition from Plaintext.

② For columnar transposition

$$\text{cipher length} = \text{rows} \times \text{columns}$$

Since $221 = 13 \times 17$, the prime number is not prime.

$$221 = 13 \times 17$$

③ Given

double transposition length = 360

$$360 = 2^3 \times 3^2 \times 5$$

④ Given

Rails = 5

length = 100

For the rail fence cipher:

cycle length

$$\text{cycle} = 2(r-1)$$

for $r = 5$

$$\text{cycle} = 2(5-1) = 8$$

Number of cycles:

$$1000 / 8 = 125$$

character distribution

Rail 1 = 125

Rail 2, 3, 4 = 125 each

⑤ In columnar transposition, character from the same column appear at intervals equal to the NO. of columns

observed periodicity

Period = 12

Thus

key length ≈ 12

$\therefore$ Hence Strong.

⑥ If the Some plaintext is encrypted using two different columnar keys

1. Each Key corresponds to a permutation of columns

2. compose ciphertext - letter position.

⑦ Key = BALLON

Letter L and O report effectively key space

$$\frac{n!}{k_1! k_2!}$$

$\therefore$ duplicate letters reduce security.

⑧ double transmission bias:

Double permutations rearranges letters twice

But

1. letter frequencies stay same.

2. Same letter pair remain near each other.

⑨ Given ciphertext length = 500

possible pairs

$$2 \leq r \leq 499$$

$O(n)$

so about 498 attempts, which is corresponding easy

⑩ Given

ciphertext length = 196

factorization

$$196 = 14 \times 14$$

possible matrix sizes

$$1 \times 196$$

$$2 \times 98$$

$$4 \times 49$$

$$14 \times 14$$

∴ thus double transposition likely cases 14-column keys

⑪ bigram distortion

Transposition keeps individual letter unchanged

so, $p(\text{letter}) = \text{constant}$

But diagrams breaks because letters move apart.

large column dept $\rightarrow$ large separation $\rightarrow$ strong diagram

⑫ Given

ciphertext length = 1200

Possible key permutation $k!$

E.g.:

$$10! = 3,628,800$$

$$15! = 1.3 \times 10^{12}$$

∴ Thus length force become impractical for larger.

⑬ let ciphertext length = N , key length = k

$$r = \lceil N/k \rceil$$

construct matrix $r \times k$

place known prefix in rows and suffix in rows.

⑭ 4-Rail fence mapping

let period $p = 6$

$$\text{Rail} = \min(i \bmod p, p - (i \bmod p))$$

Cipher index = No. of earlier characters in same rail + offset

(15) column transposition (length 255)

$$\text{Rows} = 255/15 = 15.$$

more rows $\rightarrow$ cleaner column pattern $\rightarrow$ easier frequency reconstruction.

(16) double transposition order

column permutation are non-commutative

$$\text{Generally } K_1 \circ K_2 \neq K_2 \circ K_1.$$

(17) Highly composite length (360)

many divisors $\rightarrow$ many possible key lengths —
high ambiguity.

(18) Rail fence period proof

$$\text{Period } P = 2(r-1)$$

If $i = j \pmod{P}$, zig-zag position same $\rightarrow$ same rail.

(19) Auto correlation peaks at 14, 28

likely key length = 14 (28 is multiple)

(20) double transposition (entropy), with common divisor

creates repeating blocks $\rightarrow$ structural leakage

(21) Substitution vs transposition (entropy).

Substitution entropy similar to language

Transposition same entropy as plaintext but reorder

(22) square matrix transposition

Symmetry increases patterns $\rightarrow$ reduces security

Q3 length 1024, key = 8.

Periodicity every 8 $\rightarrow$ representing statistical blocks

Q4 Rails fence with large rails

cipher = plaintext order $\rightarrow$ weak diffusion.

Q5 double transposition (7, 8)

key space $\approx 7! \times 8! \approx 1.8 \times 10^9$.

much smaller than $26!$.

Q6 In homophonic substitution one plaintext letter maps to multiple ciphertext symbols

High frequency plaintext letters are split around symbols
Thus no signal symbol shows high frequency for E.

Q7 Look for common word like "THE" or "AND".

"QEB" appears repeatedly $\rightarrow$ likely maps to "THE".

Q $\rightarrow$ T implies shift of 23 (-3).

Apply shifts to verify meaningful plaintext

Q8 pattern is ABCA (first = last)

Possible plaintext "that", "else", "never"

Given mapping A $\rightarrow$ m

B $\rightarrow$ q

C $\rightarrow$ t, Substitution must preserve repetition.

* Middle letter must differ from outer letters, limiting valid plaintext candidates

Q9 Index of coincidence (IC):

$$\sum f_i(f_i - 1)$$

$$IC = \frac{\sum f_i(f_i - 1)}{N(N-1)}$$

$$1c = 0.066, \text{ random} = 0.038$$

ciphertext, is split into $1c$ substrings.

correct key length gives $1c$ closes to English.

Small factors like 5 or 6 are most likely.

30) Kasiski examination

$$42 - 12 = 30$$

$$72 - 42 = 30$$

$$102 - 72 = 30$$

common spacing = 30

key length divide spacing

Probable key length = 2, 3, 5, 6, 10, 15

31) Hill cipher determinant = 26.

$$\gcd(\det, 26) = 1$$

$$\gcd(2, 26) = 26 \neq 1.$$

NO inverse

32) Determinant = 17

$$\gcd(17, 26) = 1$$

inverse exists

33) Diagram power

single letter probability $P(X)$

Diagram probability

$P(XY)$

Since

$$P(XY) \neq P(X)P(Y)$$

Structured preserved $\rightarrow$ Stronger constraints system.

(34) key reuse

$$C_1 = P_1 + K$$

$$C_2 = P_2 + K$$

Subtract

$$C_1 - C_2 = P_1 - P_2$$

key eliminated.

(35) double caesar

$$E(x) = x + 5$$

$$E(E(x)) = (x + 5) + 9$$

$$= x + 14$$

Equivalent Shift = 14.

(36) NO 'E' in ciphertext mapping:

$$P(E) = x$$

ciphertext letter E unused $\Rightarrow$ Some plaintext maps to E

Permutation still valid.

(37) key space

$$26! \approx 4 \times 10^{26}$$

caesar:

$$26$$

Ratio:

$$\frac{26!}{26} \approx 1.5 \times 10^{25}$$

(38) Same ciphertext
it %

$$k(P_1 - P_2) = 0 \pmod{26}$$

Non-trivial solution:

$$\det(K) = 0 \pmod{26}$$

(39) Periodicity

key length = 7

Repetition occurs at

$$\text{LCM}(7, 7) = 7$$

$$14 = 2 \times 7$$

Periodic multiple

40) matrix $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Det $3 \times 5 - 3 \times 2 = 15 - 6$
 $= 9$

$\gcd(9, 26) = 1$

41) $IC \approx 0.066$

English $IC = 0.065$

So

$K = 1$

42) $T \rightarrow Q$ Z

$T \rightarrow Q$

$H \rightarrow Z$

Constraints

No other letter map to Q or Z.

43) let key

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$H(7, 8) \rightarrow T(19, 2)$

$7a + 8b = 19$

$7c + 8d = 2$

$B(1, 24) \rightarrow K(10, 12)$

$a + 2b = 10$

$c + 2d = 12$

Solve modulo 26.

44) Repetition every 13

Distance = 13

key length divides 13

$K = 13$

45) Vigenere algebra

$C_i = P_i + K_i \text{ mod } m$

for key:

$K = (10, 4, 24)$

Equivalent to:

Shift 10, Shift 4, Shift 24 cyclically.

(46) If k is fixed:

$$(26-k)!$$

Search Space reduced

(47) if:

$$\gcd(\det, 26) \neq 1.$$

Then:

$\det^{-1}$ does not exist

(48) large key

As key length $\rightarrow N$

$$1c \rightarrow 0.038$$

Distribution $\rightarrow$ uniform

(49) If:

$$k_1 \equiv k_2 \pmod{26}$$

Then

$$P + K_1 = P + K_2$$

Equivalent keys.

(50) Affine cipher

$$H=7 \rightarrow Z=25$$

$$E=4 \rightarrow E=4$$

$$\text{Equations: } 25 = 7a + b$$

$$4 = 4a + b$$

$$\text{Subtract: } 21 = 3a$$

$$a = 7.$$

$$\text{Substitute: } 4 = 28 + b$$

$$4 = 2 + b$$

$$b = 2.$$

$$a = 7, b = 2.$$

$$\gcd(7, 26) = 1.$$

✓ Good

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